HISTOLOGY OF MUSCLES

LEARNING OBJECTIVES

By the end of this lecture, the students should be able to understand:

- three types of muscle at the light and electron microscope levels, including distinctive features of each muscle fiber.
- Describe the structural basis of muscle striations.
- Know the structural elements that produces muscle contraction and brings the movement of a body part.
- Know the function and organization of the connective tissue in muscle.

MUSCLE FUNCTION

1. Contraction for locomotion and skeletal movement.
2. Contraction for propulsion.
3. Contraction for pressure regulation.

Muscle classification

- Muscle tissue may be classified according to a morphological classification or a functional classification.
- There are two types of muscle based on the morphological classification system
  1. **Striated**
  2. **Non striated or smooth.**

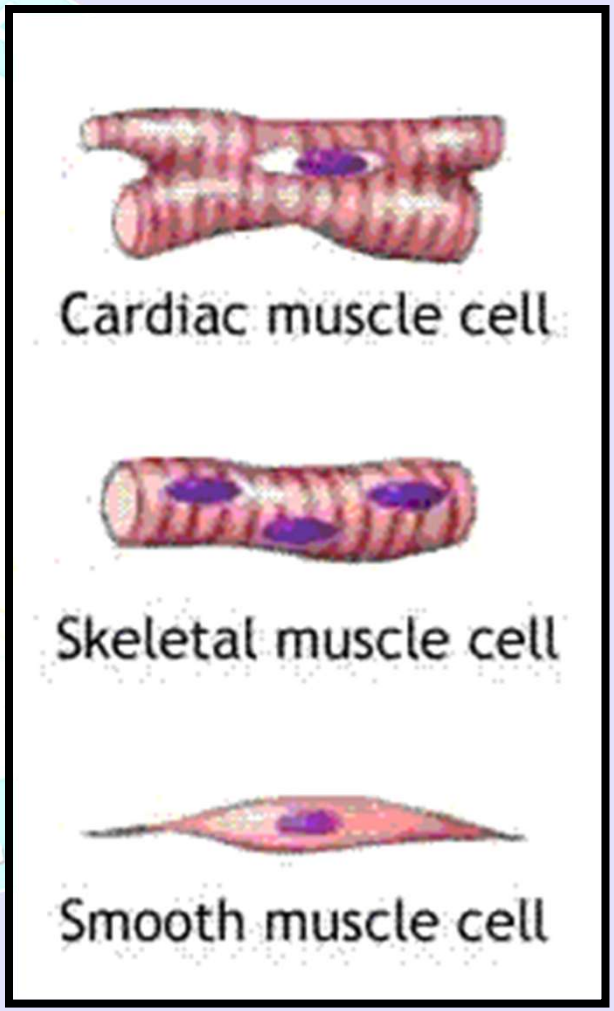
Functional classification

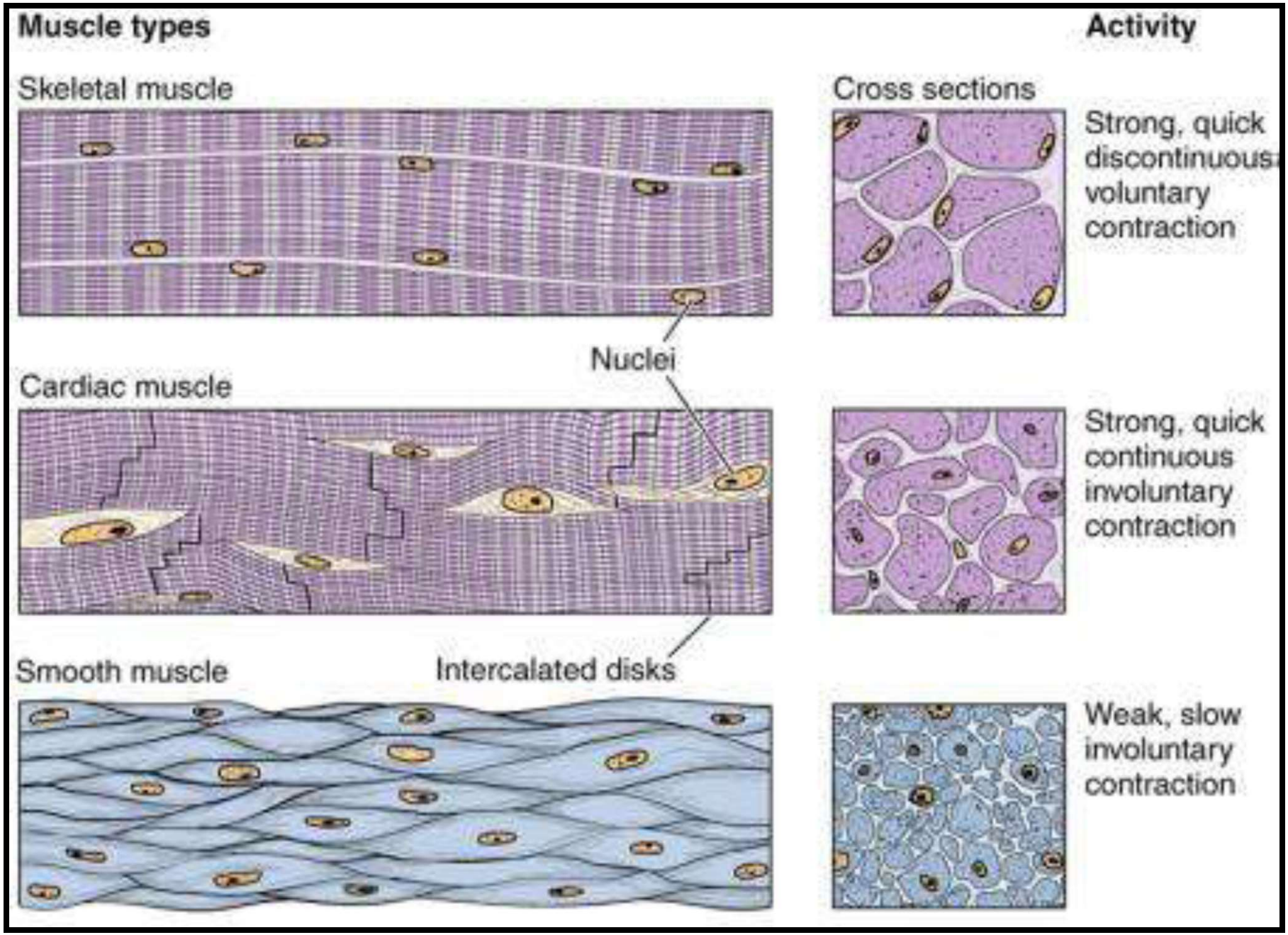
- There are two types of muscle based on a functional classification system
  1. **Voluntary**
  2. **Involuntary.**

TYPES OF MUSCLE

- Types of muscle: there are generally considered to be three types of muscle in the human body.

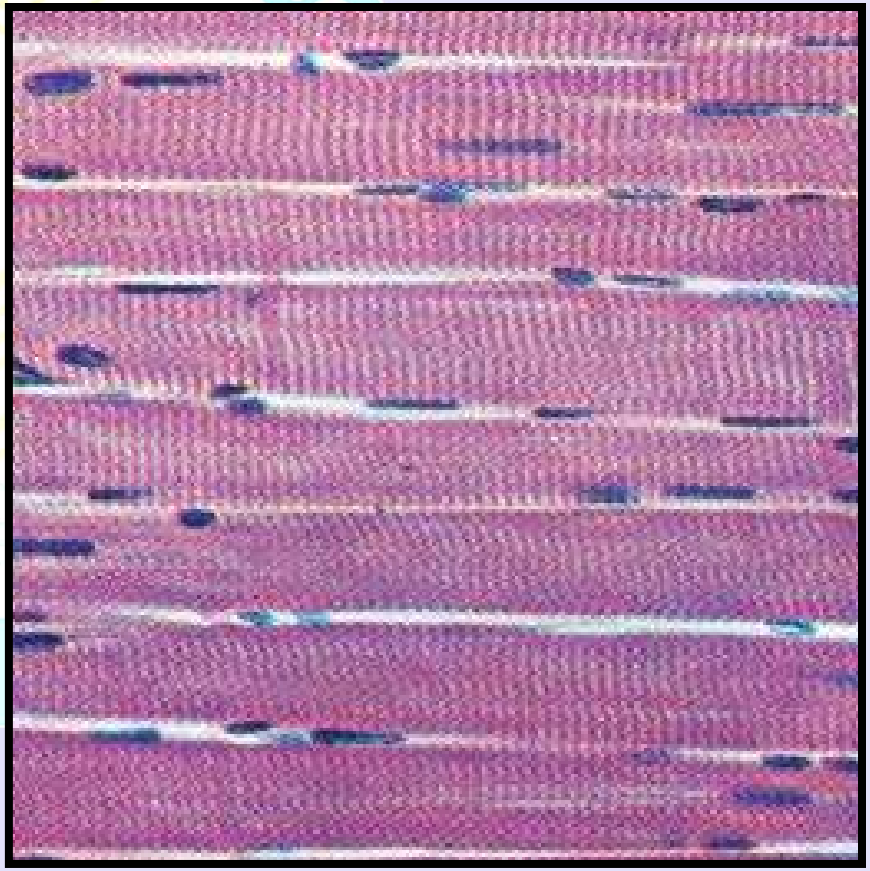
1. ***Skeletal muscle:*** which is striated and voluntary
2. ***Cardiac muscle:*** which is striated and involuntary
3. ***Smooth muscle:*** which is non striated and involuntary




Muscle types
Activity
Skeletal muscle
Cross sections
Strong, quick discontinuous voluntary contraction
Nuclei
Cardiac muscle
Strong, quick continuous involuntary contraction
Intercalated disks
Smooth muscle
Weak, slow involuntary contraction

CHARACTERISTICS OF SKELETAL MUSCLE

1. Skeletal muscle cells are elongated or tubular.
2. multiple nuclei, located on the periphery of the cell.
3. Skeletal muscle is striated.



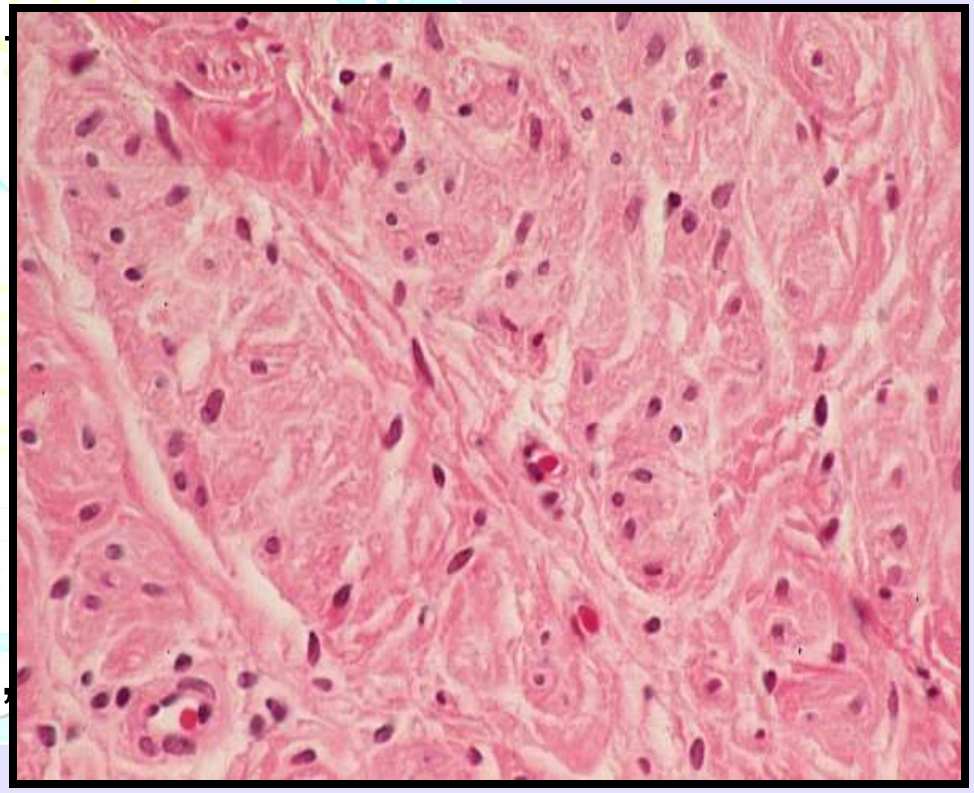
CHARACTERISTICS OF CARDIAC MUSCLE

1. Cardiac muscle cells are not as long as skeletal muscles cells and often are branched.
2. Cardiac muscle cells may be mononucleated or binucleated.
3. nuclei are located centrally in the cell.
4. Cardiac muscle is also striated.
5. In addition cardiac muscle contains intercalated discs.



CHARACTERISTICS OF SMOOTH MUSCLE

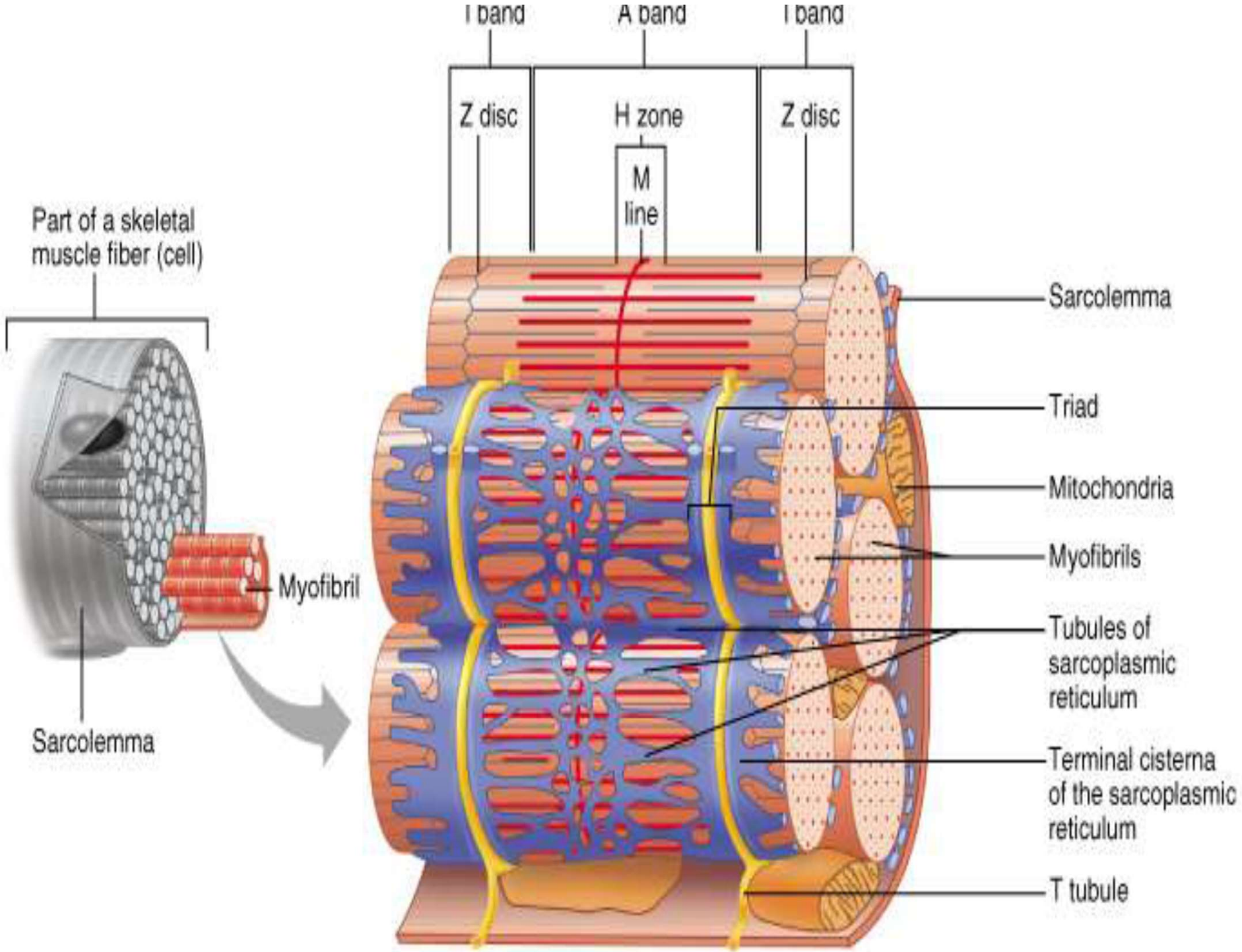
- Smooth muscle cell are described as spindle shaped. Wide in the middle and narrow to almost a point at both ends.
- Single centrally located nucleus.
- Smooth muscle cells do not have visible striations although they do
- Contains the same contractile proteins as skeletal and cardiac muscle, but laid out in a different pattern.



MUSCLE TERMINOLOGY

Myofiber Or Myocyte: a muscle cell

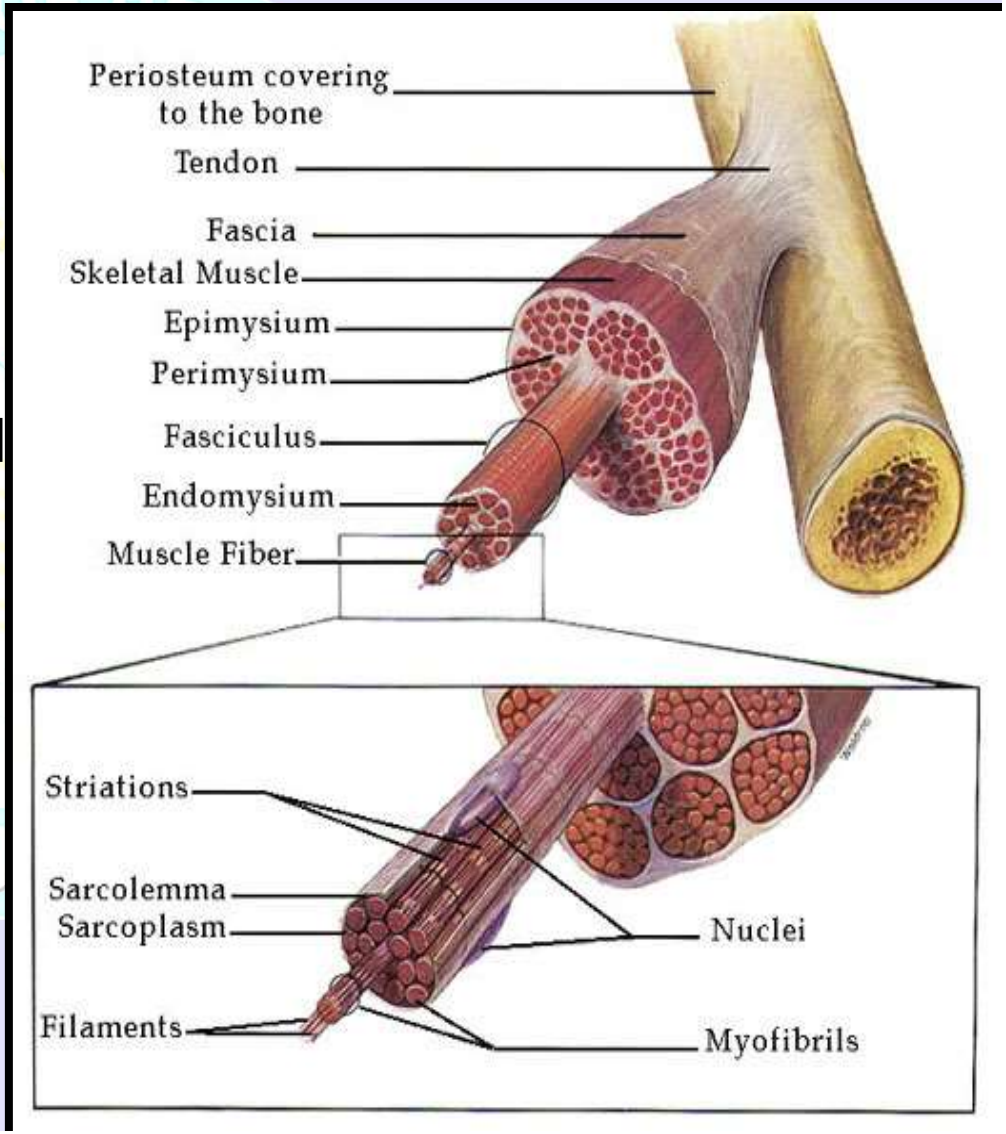
1. **Sarcolemma:** the plasma membrane of a muscle cell
2. **Sarcoplasm**: the cytoplasm of the muscle cell
3. **Sarcoplasmic** reticulum: the endoplasmic reticulum of a muscle cell
4. **Sarcosome**: the mitochondria of a muscle cell
5. **Sarcomere**: the contractile or functional unit of muscle

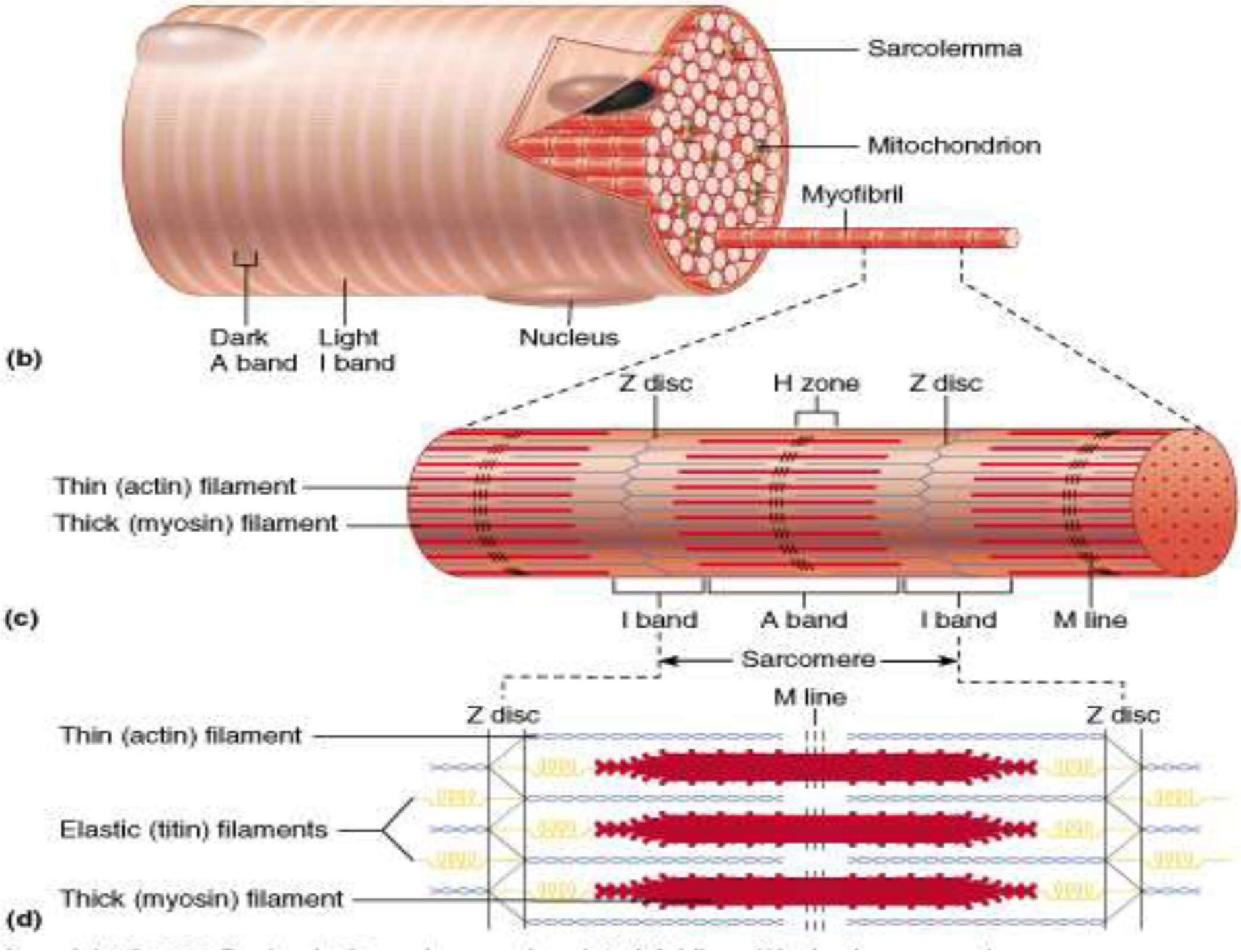

I band
A band
I band
Z disc
H zone
Z disc
M line
Part of a skeletal muscle fiber (cell)
Myofibril
Sarcolemma
Sarcolemma
Triad
Mitochondria
Myofibrils
Tubules of sarcoplasmic reticulum
Terminal cisterna of the sarcoplasmic reticulum
T tubule

- There are specialized invaginations of the sarcolemma that run transversely across the cell known as T tubules (transverse tubules).
- Terminal Cisternae are sac-like regions of the sarcoplasmic reticulum.
- The terminal cisternae act as calcium storage sites. The calcium ions stored in the terminal cisternae are essential in muscle contraction.

MICROSCOPIC STRUCTURE OF SKELETAL MUSCLE

- The entire muscle is surrounded by a connective tissue called the **epimysium**.
- Fascicles are bundles of individual muscle cells (myofibers or myocytes).
- These bundles are surrounded by a connective tissue sheath called the **perimysium**.
- Each fascicle is made up of several muscle cells known as myocytes or myofibers or delicate muscle fibers.
- Each muscle Cell is surrounded by a connective tissue sheath known as the **endomysium**

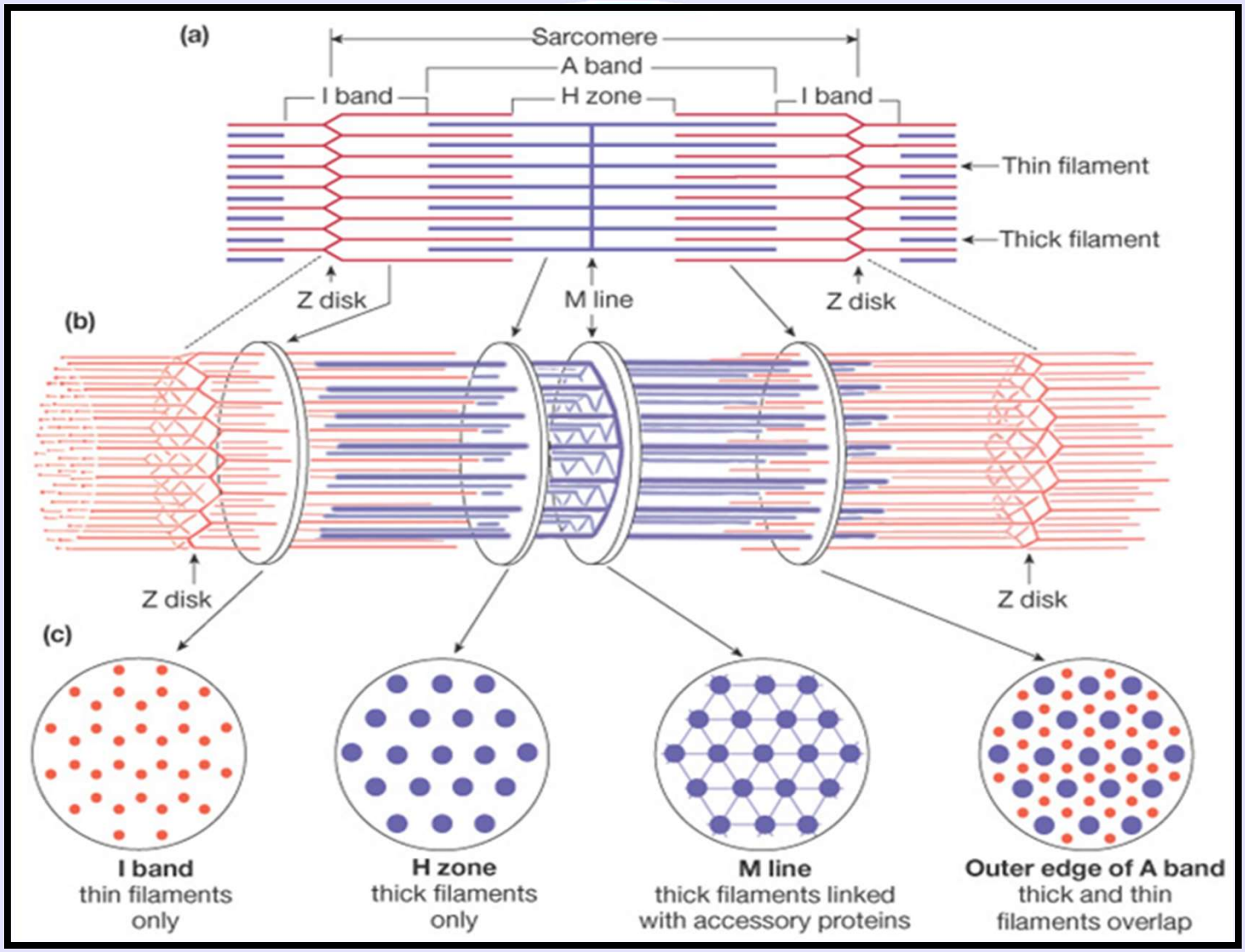



Sarcolemma
Mitochondrion
Myofibril
Dark
A band
Light
I band
Nucleus
(b)
Z disc
H zone
Z disc
Thin (actin) filament
Thick (myosin) filament
(c)
I band
A band
I band
M line
Sarcomere
M line
Z disc
Z disc
Thin (actin) filament
Elastic (titin) filaments
Thick (myosin) filament
(d)

ORGANIZATION OF SKELETAL MUSCLE FIBERS

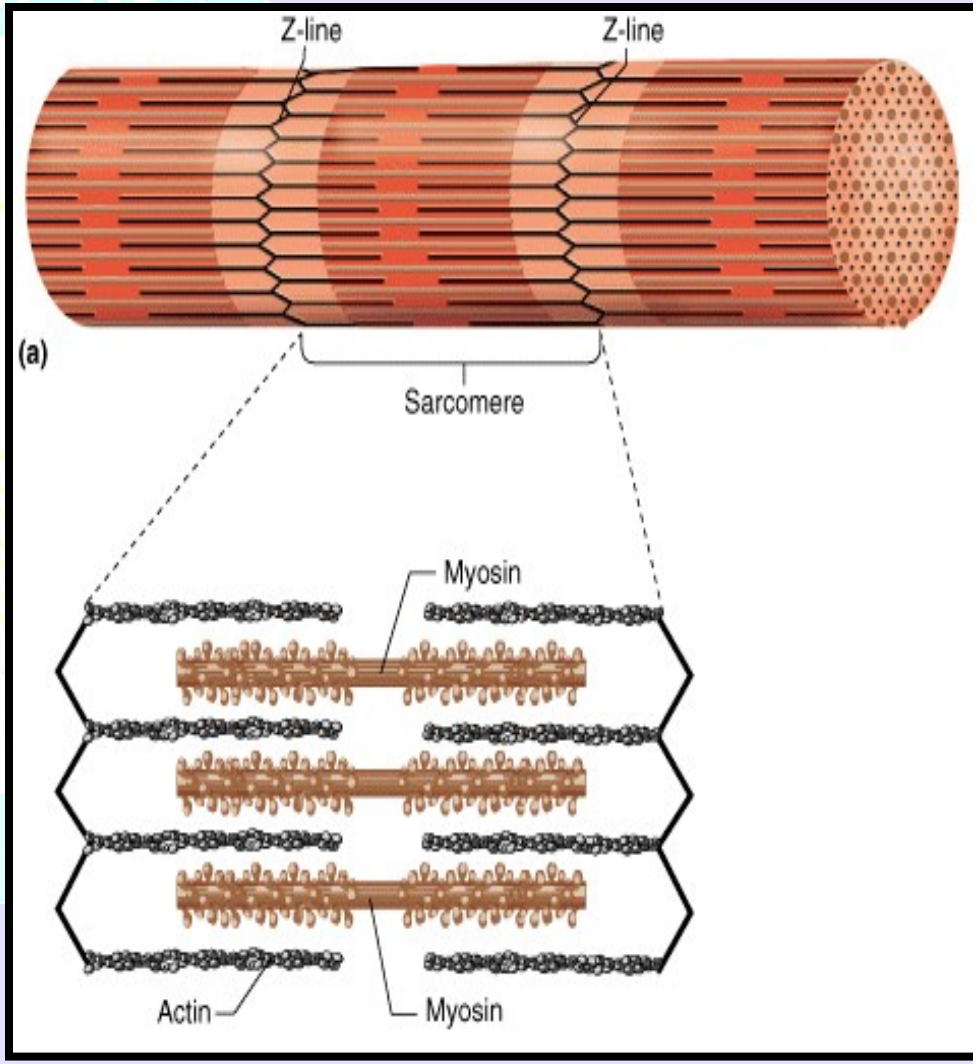
- The sarcomere is the repeating unit of a myofibril in a muscle cell, composed of an array of overlapping thick and thin filaments between two adjacent Z discs.
- The muscle fibers show the cross-section of alternating light and dark bands. The dark bands are called A bands and light bands are called Ibands.
- In electron microscope, each I band is bisected by a dark transverse Z line.
- The sarcomere extends from Z line to Z line and is about 2.5μm in resting muscles.

SARCOMERE


(a)
Sarcomere
A band
I band
H zone
I band
Thin filament
Thick filament
Z disk
M line
Z disk
(b)
Z disk
Z disk
(c)
I band
thin filaments
only
H zone
thick filaments
only
M line
thick filaments linked
with accessory proteins
Outer edge of A band
thick and thin
filaments overlap

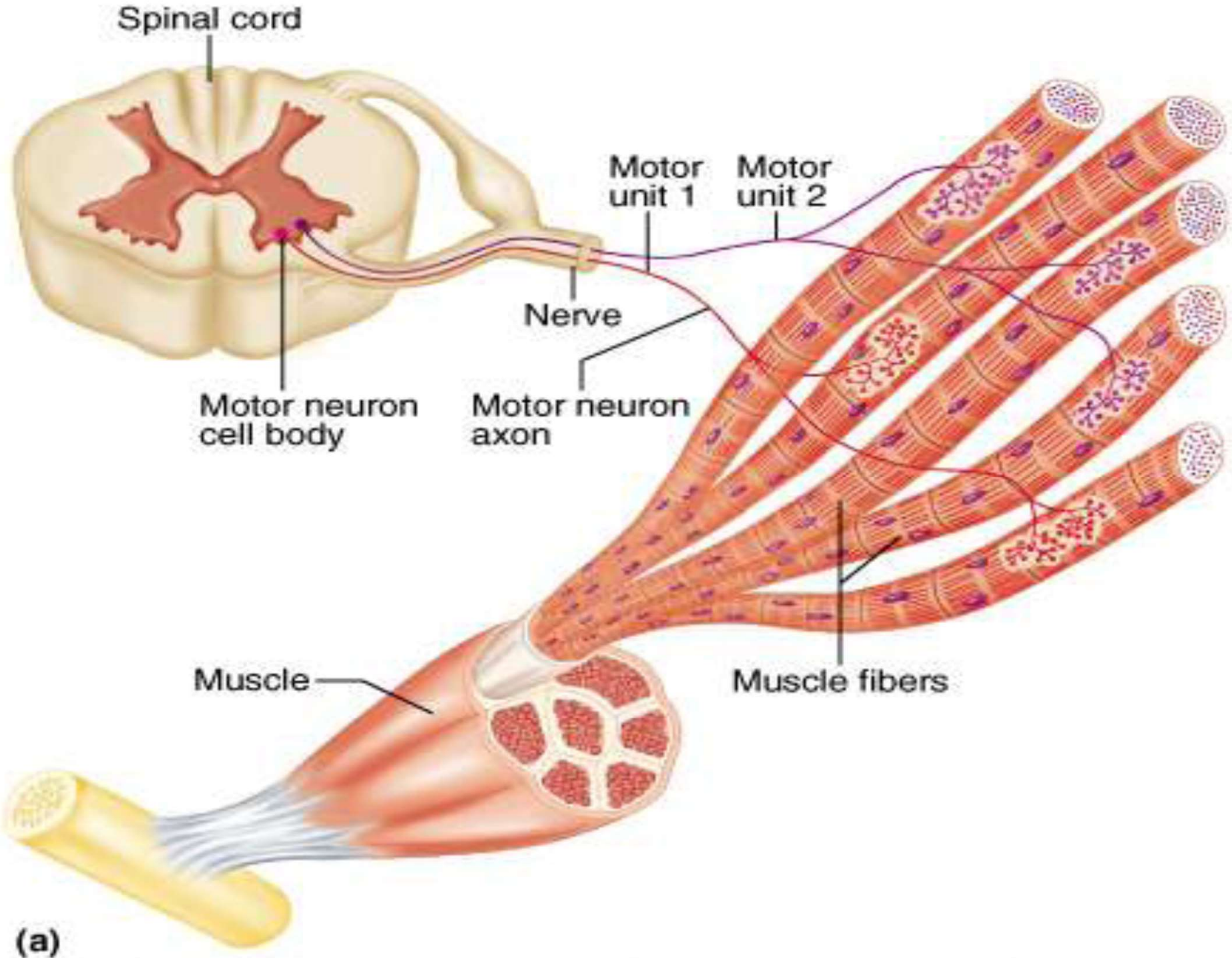
THICK AND THIN FILAMENTS

- Sarcomere pattern is due to two types of filaments, thick and thin.
- Thick filaments occupy A band.
- The I band consists of thin filaments.
- The thick A band overlap the thin I band filaments
- The A band shows the presence of light zones in center, the H band containing only Myosin molecules.
- The H band is bisected by „M" line.



BLOOD SUPPLY OF SKELETAL MUSCLES

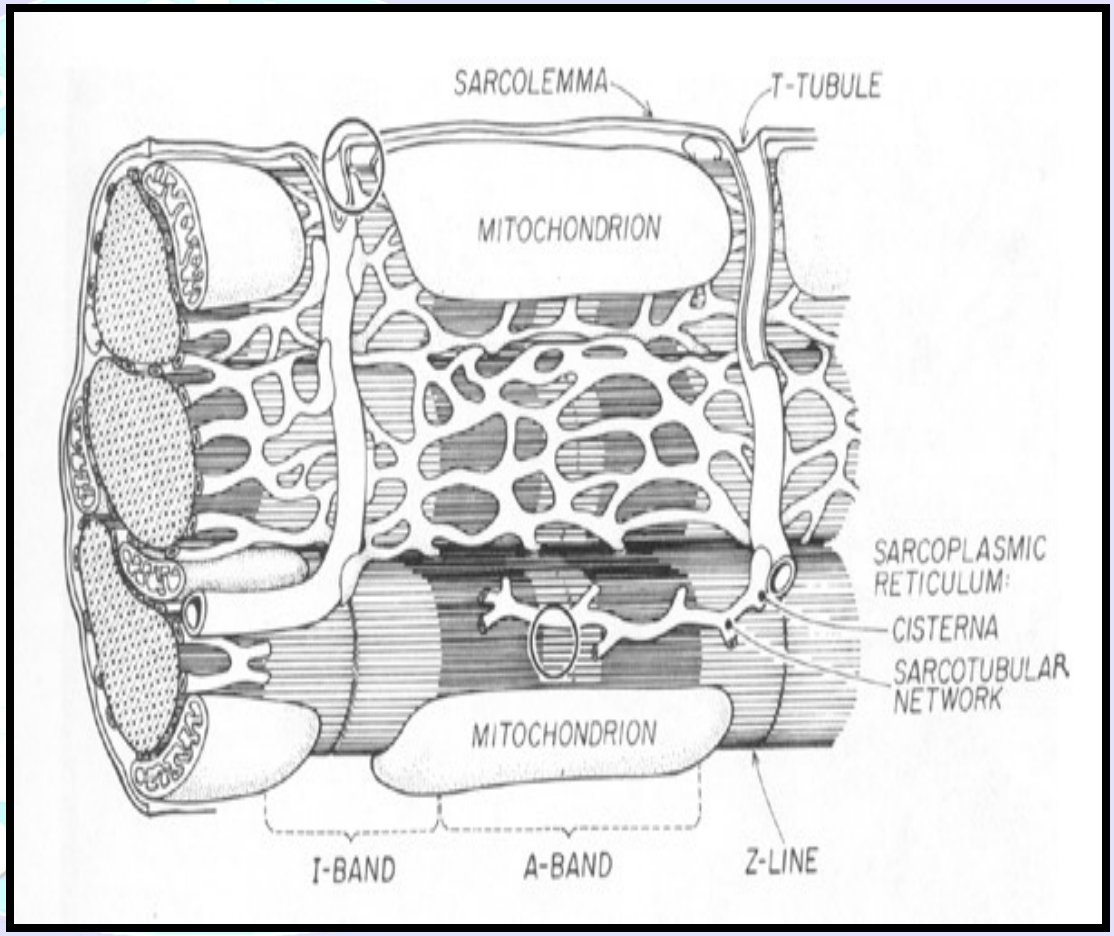
- Rich capillary supply.
- Nerve supply.
- One motor nerve ending form a somatic nerve.
- One nerve fiber innervate a single muscle fiber.
- A motor nerve fiber and muscle fiber are collectively known as a **motor unit.**


Spinal cord
Motor unit 1
Motor unit 2
Nerve
Motor neuron cell body
Motor neuron axon
Muscle
Muscle fibers
(a)

CARDIAC MUSCLES

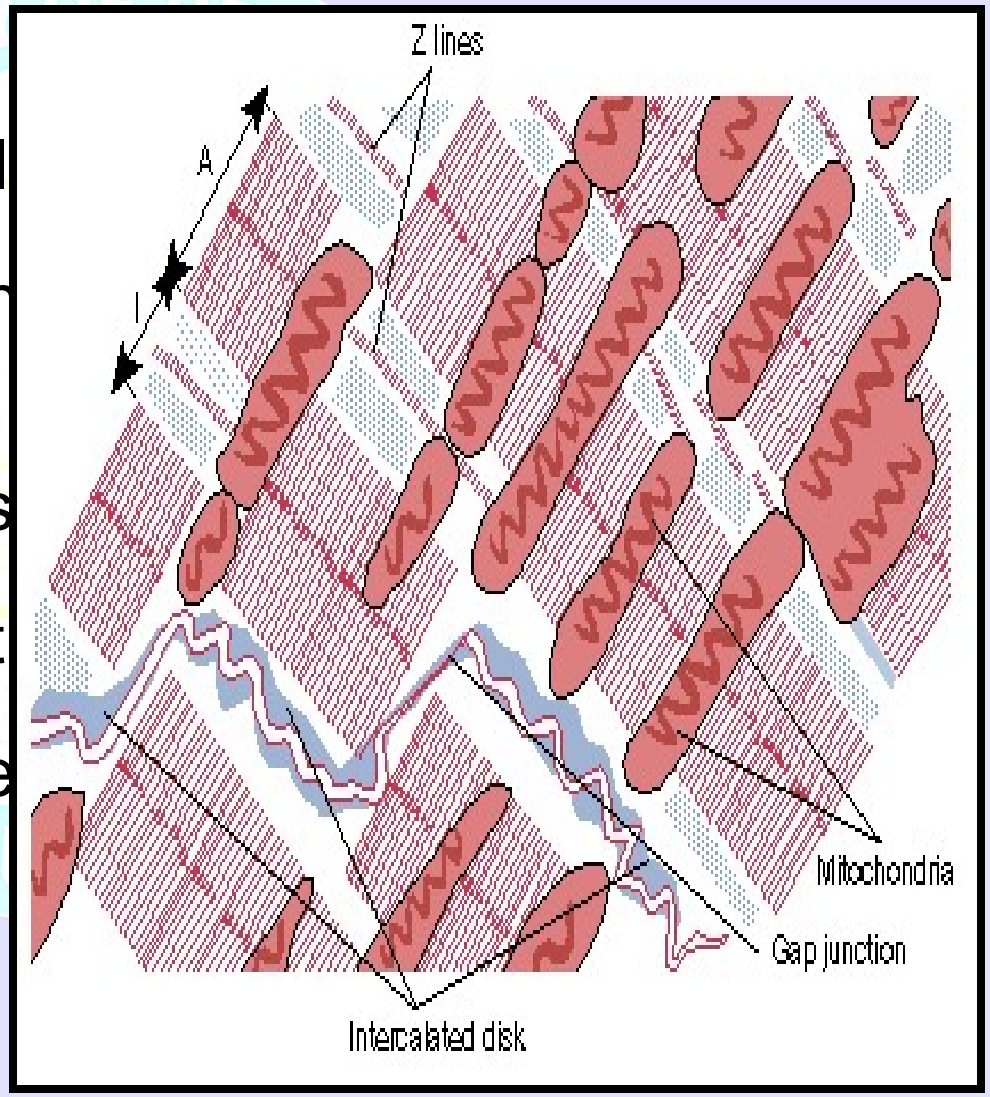
1. Involuntary .
2. Striated,
3. Nucleus is oval.
4. Found in Myocardium.
5. Present in the form of chains.
6. Cells within each chain bifurcate or branch giving an appearance of syncytial network.
7. Unique character of cardiac muscle is presence of darkly stained transverse lines that cross the chain of cardiac cells, these bands often follow an irregular step like course and are known as **Intercalated discs**.

- The T-tubule are large in ventricular muscles..
- Cardiac t-tubule are found at the level of „Z" band, rather than A-I junction.
- The sarcoplasmic reticulum is not well developed.
- Triads are not common in cardiac muscles but T-tubule is associated with one lateral endoplasmic reticulum cisternae are called **Diads**.
- In old age, lypofuscin pigment is deposited around nucleus.



SMOOTH MUSCLES

1. Smooth muscle is found in stomach, intestine, urinary and genital tract.
2. Fusiform cells.
3. Striations are longitudinal only.
4. Longitudinal striations run throughout length.
5. Single central nucleus.
6. Sarcoplasmic reticulum is rudimentary.
7. T-tubules are not present
8. Characteristic contractile activity depends upon the Actin and Myosin filaments.
9. There is little connecting tissue b/w adjacent myofibrils.



THE END